

July 7, 2025

Ben Kochman, Acting Administrator
U.S. Department of Transportation
Pipeline and Hazardous Materials Safety Administration
1200 New Jersey Avenue, SE Washington, DC 20590

RE: public comment on ANPRM – Pipeline Safety: Amendments to Liquefied Natural Gas Facilities (Docket No. PHMSA-2019-0091), 90 Fed. Reg. 18,949 (May 5, 2025)

Dear Acting Administrator Kochman:

Please accept these comments submitted on behalf of the Center for Biological Diversity, Food & Water Watch, and the Science and Environmental Health Network. We appreciate the opportunity to comment on these essential regulations, which have not been updated in more than 20 years, to ensure the safety of all pipeline infrastructure.

Our comments below focus on the intersection of liquefied natural gas (“LNG”) pipeline and carbon dioxide (“CO₂”) pipeline safety. It is critically important for the agency to consider the unique safety risks and vulnerabilities that co-location of LNG and CO₂ pipelines may present, particularly as project developers increasingly plan to deploy carbon capture and storage/sequestration (“CCS”) equipment at LNG facilities, as well as safety vulnerabilities associated with reuse of existing, often aged, LNG pipelines for CO₂ transport.

Vulnerabilities of Current CO₂ Pipeline Safety Regulations

The Pipeline Safety Trust explains that under current safety regulations, “[t]he Pipeline and Hazardous Materials Safety Administration (“PHMSA”) currently exercises no jurisdiction over pipelines transporting CO₂ as a gas or liquid, and only regulates CO₂ pipelines with a concentration of more than 90% carbon dioxide compressed to a supercritical state, rendering any pipeline moving CO₂ in any other state or with less than 90% purity entirely unregulated by the federal pipeline safety agency. There are other large regulatory gaps around siting, fracture mitigation, determining potential impact areas, use of odorant, emergency response, and contaminants.”¹

For example, CO₂ has unique safety hazards not contemplated by current regulations—it is a toxicant and asphyxiant heavier than air, and has a potential impact area measured in miles, not feet like with natural gas pipelines. Further, CO₂ creates a unique fracture potential for pipelines, and releases can result in very violent ruptures with the “unzipping” of a pipeline over long distances. These risks are inherent in transporting pressurized supercritical CO₂, which becomes

¹ Pipeline Safety Trust (March 2022), <https://pstrust.org/wp-content/uploads/2022/03/CO2-Pipeline-Backgrounder-Final.pdf> (emphasis added); see also Accufacts, *Perspectives on the State of Federal Carbon Dioxide Transmission Pipeline Safety Regulations as it Relates to Carbon Capture, Utilization, and Sequestration within the U.S.* (March 23, 2022), <https://pstrust.org/wp-content/uploads/2022/03/3-23-22-Final-Accufacts-CO2-Pipeline-Report2.pdf>.

highly corrosive in the presence of water, creating carbonic acid, which can corrode and embrittle pipelines.

The current PHMSA regulations very narrowly define the type of CO₂ covered; they only regulate CO₂ pipelines if transported as a supercritical fluid of >90% purity. If CO₂ is being transported as a gas, or a liquid, or at <90% purity, it is entirely unregulated. Additionally, these regulations do not include a requirement for odorants to be added (like with natural gas) for leak detection safety. The regulations also include no standards on contaminants within transported CO₂; however, it is common for transported CO₂ to contain toxic and corrosive contaminants.

The inadequacy of PHMSA's existing CO₂ pipeline safety regulations was exposed by the February 2020 CO₂ pipeline rupture in Satartia, Mississippi which sent dozens of community members to the hospital and demonstrated the significant and severe harms that can befall a community nearly a mile from a CO₂ pipeline rupture site.² This incident particularly highlighted the failure of existing regulations to require, and for the operator to provide, adequate notice of any pipeline incident to the community as well as training for first responders on unique risks of CO₂ releases.

High concentrations of CO₂, such as that experienced during the Satartia incident, can displace oxygen in the air and cause combustion engines to stall, hindering evacuations and emergency response efforts. In Satartia, first responders were stymied in their efforts to aid community members suffering from asphyxiation from the pipeline rupture and CO₂ release by emergency vehicles repeatedly stalling and rendered inoperable.³ First responders in Satartia described the scene as “almost like something you’d see in a zombie movie,” with community members walking dazed in circles, disoriented, and passed out unresponsive.⁴

Scientific American has reported that “If a person [in the vicinity of a CO₂ pipeline rupture] breathes air with a 10 percent concentration of CO₂, they can fall unconscious within one minute.”⁵ The National Institute for Occupational Safety and Health considers exposure to 4% concentrations of CO₂ to be “immediately dangerous to life and health,” meaning that if a person is exposed to 4% CO₂ for just 30 minutes they may become sufficiently disoriented that they are unable to get to safety.⁶ Exposure to CO₂ can also lead to brain damage from lack of oxygen (hypoxia), as well as chronic disabilities from CO₂-induced acidification of major organs in the

² Dan Zegart, Huffington Post, “The Gassing of Satartia” (Aug. 26, 2021), https://www.huffpost.com/entry/gassing-satartia-mississippi-co2-pipeline_n_60ddea9fe4b0ddef8b0ddc8f.

³ “The Gassing of Satartia,” *supra*.

⁴ *Id.*

⁵ Anna Matson, Scientific American, *Pipelines Touted as Carbon Capture Solution Spark Uncertainty and Opposition*, <https://www.scientificamerican.com/article/pipelines-touted-as-carbon-capture-solution-spark-uncertainty-and-opposition/>.

⁶ Nat’l Inst. for Occupational Safety and Health, <https://www.cdc.gov/niosh/idlh/124389.html>; see also Kris Permentier et al., *Carbon dioxide poisoning: a literature review of an often forgotten cause of intoxication in the emergency department*, International Journal of Emergency Medicine Vol. 10, No. 14 (2017), <https://bit.ly/3umxqNv>; U.S. Dep’t of Agriculture, Food Safety and Inspection Service Environmental, Safety and Health Group, *Carbon Dioxide Health Hazard Information Sheet*, <https://bit.ly/2VCriqC>; “Carbon Dioxide,” in NIOSH Pocket Guide to Chemical Hazards, Centers for Disease Control and Prevention (Oct. 30, 2019), <https://bit.ly/3tyCGON>.

body that can permanently damage lungs, liver, and kidneys. One survivor of the Satartia incident described the lingering adverse effects of spending more than an hour unconscious in a car filled with CO₂ during the Satartia pipeline rupture which, three and a half years later, still render him too disabled to work.⁷

After finalizing its incident report and investigation into CO₂ pipeline operator Denbury's failures which led to the rupture,⁸ PHMSA announced its intention to update its CO₂ pipeline safety regulations in direct response to the Satartia incident.⁹ But, still lacking adequate safety rules, we saw another CO₂ pipeline leak in April of 2024 in Sulphur, Louisiana (also operated by Denbury, now owned by Exxon), which again endangered numerous residents and saw significant confusion among first responders.¹⁰

In response to these acknowledged regulatory safety inadequacies, PHMSA had developed updated safety standards that were made publicly available in January of 2025 but were rescinded prior to being published in the Federal Register for public comment.¹¹ Thus, the current operative CO₂ pipeline safety regulations still fail to adequately protect public health and safety.

Co-Location of LNG and CO₂ Pipelines

Given the above-discussed inadequacy of PHMSA's current CO₂ pipeline safety regulations, as well as the unique properties of CO₂, especially when liquified and transported at high pressure, it is critically important for the agency to consider the safety vulnerabilities inherent in co-locating LNG and CO₂ pipeline infrastructure in this rulemaking. The numerous safety concerns with CO₂ pipelines, currently un- and under-regulated, as described above, increase vulnerability for any co-located LNG pipeline(s) because of potential domino effects: a failure in any one pipeline would likely trigger failures in adjacent pipelines, significantly increasing overall risk. Additionally, ruptures, leaks or releases involving both types of pipelines could result in significantly more severe consequences than a single incident, with larger affected area, more confusion for first-responders, and greater risk to the safety of workers and communities.

⁷ Julia Simon, *The U.S. is expanding CO2 pipelines. One poisoned town want you to know its story*, National Public Radio (May 21, 2023), <https://www.npr.org/2023/05/21/1172679786/carbon-capture-carbon-dioxide-pipeline>.

⁸ See Pipeline and Hazardous Materials Safety Admin., *Failure Investigation Report: Denbury Gulf Coast Pipelines LLC* (May 26, 2022), <https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/2022-05/Failure%20Investigation%20Report%20-%20Denbury%20Gulf%20Coast%20Pipeline.pdf>.

⁹ Pipeline and Hazardous Materials Safety Admin., *PHMSA Announces New Safety Measures to Protect Americans From Carbon Dioxide Pipeline Failures After Satartia, MS Leak* (May 26, 2022), <https://www.phmsa.dot.gov/news/phmsa-announces-new-safety-measures-protect-americans-carbon-dioxide-pipeline-failures>.

¹⁰ See Video: CO2 Pipeline Leak, Sulphur, Louisiana Apr. 3, 2024, <http://vimeo.com/showcase/11130725> (hear local people tell their stories about the CO2 pipeline leak that spewed dangerous CO2 for two hours before the owner (Denbury/ExxonMobil) shut it down); see also Pipeline Safety Trust, *Denbury, Pipeline Company Responsible for 2020 Carbon Dioxide Pipeline Rupture in Satartia, MS Leaks CO2 Again, This Time in Sulphur, LA* (Apr. 5, 2024), <https://pstrust.org/denbury-pipeline-company-responsible-for-2020-carbon-dioxide-pipeline-rupture-in-satartia-ms-leaks-co2-again-this-time-in-sulphur-la/>.

¹¹ See Pipeline and Hazardous Materials Safety Admin., *Notice of Proposed Rulemaking* (Jan. 10, 2025) <https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/2025-01/PHMSA%20Notice%20of%20Proposed%20Rulemaking%20for%20CO2%20Pipelines%20-%202137-AF60.pdf>.

Reuse of LNG Pipelines for CO₂

New projects seeking to convert former LNG pipelines for use with CO₂ present huge unknown risks, which are also currently under-regulated. Existing guidance for converting existing LNG pipelines into primarily CO₂ gas service are inadequate and non-binding.¹² The operative Advisory Bulletin on such potential conversion “does not carry the force of promulgated pipeline safety regulation” and “does not address, nor was it intended to address, the specific additional challenges associated with unique fracture propagation risks associated with CO₂ transmission pipelines.”¹³

In an report commissioned by the Pipeline Safety Trust, Richard Kuprewicz of Accufacts opined that “[s]uch pipeline conversions would be at much greater risk of failure from CO₂ service than conventional hydrocarbon or new construction CO₂ pipelines, given the unique and increased potential for CO₂ pipeline ruptures from various risks associated with CO₂ operation.”¹⁴ Not only because such existing LNG pipeline infrastructure is likely significantly aged, many of the unique characteristics of CO₂ itself present significant safety risks and unknowns. For example, given potential for corrosion and catastrophic breaching in the case of CO₂, as discussed above, for CO₂ pipelines specifications include requirements for stainless steel linings and other anti-corrosion measures not found or required in gas and oil pipelines.

Conclusion

The Advanced Notice of Proposed Rulemaking itself “[r]ecogniz[es] that PHMSA regulations have not kept pace with technological innovation and best practices in the LNG industry,”¹⁵ and so too have they failed to address and keep pace with the boom in CCS project development and associated CO₂ pipeline infrastructure across the United States. For these reasons, we respectfully request that the agency carefully consider and address the significant safety risks that arise from the co-location or reuse of LNG pipelines with those transporting CO₂.

With any questions concerning this comment letter, please contact:

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Respectfully submitted,

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¹² See Pipeline and Hazardous Materials Safety Admin., *Advisory Bulletin, ADB-2014—04, Pipeline Safety: Guidance for Pipeline Flow Reversals, Product Changes and Conversion to Service*, Docket No. PHMSA–2014–0040 (Sept 12, 2014).

¹³ See Accufacts, *supra* n.1.

¹⁴ *Id.*

¹⁵ 90 Fed. Reg. at 18,950.